

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Supported the **Linux Yocto build process**, including customization of the kernel and file system, and application of necessary patches for a 5,000-node device fleet.
- Contributed to the maintenance and development of **container-based build environments** using Docker, integrating with Jenkins CI/CD pipelines.
- Developed and maintained module requirements, design specifications, and module test plans for embedded software components, working with **SRS Engineer** and senior developers.
- Executed testing at **black box, integration, and unit levels** for embedded firmware, ensuring adherence to defined test cases.
- Developed source code adhering to **coding guidelines** and MISRA standards, performing static code analysis using Klocwork to identify potential issues.
- Organized formal **code and document reviews** of own work products, ensuring compliance with company QA requirements and ASPICE standards.

SKILLS

Embedded Systems – Firmware, Drivers, Middleware, Low-level software, Hardware-Software Interface, GPIOs, SPI, I2C, UART, Ethernet, Memory.

Development Tools – Linux, Yocto, Kernel Customization, File System Customization, Patches, Docker, Containerization, Compiler, Linker, CMake, Klocwork, Codebeamer.

Programming Languages – C, C++, Python, Java, Spring Boot.

Software Practices – Agile, Scrum, CI/CD, Jenkins, Git, Code Reviews, MISRA Standard, ASPICE, ISO15504, QA Requirements, Software Quality Processes.

Cloud & Scripting – AWS (EC2, S3, Lambda, CloudWatch), Python Scripting, Shell Scripting.

Data & Documentation – SQL, PostgreSQL, MySQL, Oracle, Jira, Confluence, Wiki-page documentation, Google Site design.

PROJECTS

Embedded Linux Platform Development for Automotive SoC

- Built a **vision framework** for a new Texas Instruments SoC device, enabling development and demonstration on ECU hardware.
- Supported the **Linux Yocto build process** for the platform, customizing the kernel and file system, and integrating with **container-based build environments**.
- Adapted the toolchain and set up project repositories according to the vision platform SW standards and **Valeo standards**.

AI-Driven Code Generation for Platform Deployment

- Developed an end-to-end **AI solution** to automate requirement analysis and code generation for platform deployment into customer projects.
- Utilized Python scripting for data pipeline and ETL processes, generating code artifacts for embedded systems.
- Integrated with **source code management** tools and performed static code analysis to ensure quality and compliance.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)